

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-YD-2301-01 Sheet 1 of 1
OPL Title	Nitrogen Inerting Before Steam-On	Discipline	Process / Operations	
Equipment	YD-2301 - POLYMER FLUID BED DRYER	Area / Unit	2300 - POWDER DRYING SYSTEM	
Related Interlock	SEQ-5500 (SIL 2)	P&ID; Ref	TJC-LLD-PID-2301	
Classification:	☒ Basic Knowledge	☐ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **nitrogen inerting before steam-on** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. Polymer powder + oxygen + heat can cause a dust explosion. The dryer must be fully inerted before any steam is admitted. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Start the N2 purge and raise FT-2302 flow above 200 kg/h	Permit valid & briefed
2	Confirm the vent O2 analyzer AT-2307 reads below 5 %	Parameter within limit
3	Only then admit steam gradually via TIC-2301 / TV-2301	Interlock status confirmed
4	Never bypass the FSLL-2302 low-flow inert trip to start faster	Change logged on DCS
5	If O2 rises above 8 %, the ASHH-2307 trip will stop steam and drive	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
Scheduled proof test of dryer outlet high-high temperature trip	Set-point within tolerance, minor zero drift corrected	Recalibrated TSHH-2301 to 125 degC and function tested to logic

6. KEY LEARNING POINTS

- Polymer powder + oxygen + heat can cause a dust explosion. The dryer must be fully inerted before any steam is admitted.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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Cross-ref tag YD-2301 across Datasheet, GA, P&ID; Interlock and Maintenance History.